

ALFIDO TECH INTERNSHIP

TASK 2 – PWM MOTOR CONTROL WITH ULTRASONIC OBSTACLE DETECTION

Submitted by: Aditya Raj | Submission Date: 1 September 2026

Task 2: Control DC Motor Speed using PWM and Stop when Ultrasonic Sensor Detects an Obstacle

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```
8 void setup() {
13   Serial.begin(9600);
14 }
15
16 void loop() {
17   digitalWrite(TRIG_PIN, LOW);
18   delayMicroseconds(2);
19
20   digitalWrite(TRIG_PIN, HIGH);
21   delayMicroseconds(10);
22   digitalWrite(TRIG_PIN, LOW);
23
24   long duration = pulseIn(ECHO_PIN, HIGH, 30000);
25
26   float distance = duration * 0.0343 / 2;
27
28   Serial.print("Distance: ");
29   Serial.print(distance);
30   Serial.println(" cm");
31
32   if (distance <= stopDistance) {
33     digitalWrite(LED_PIN, 0);
34     Serial.println("OBSTACLE - MOTOR STOPPED");
35   }
36   else {
37     digitalWrite(LED_PIN, motorSpeed);
38     Serial.println("CLEAR - MOTOR RUNNING");
39   }
40
41   delay(200);
42 }
```

Simulation

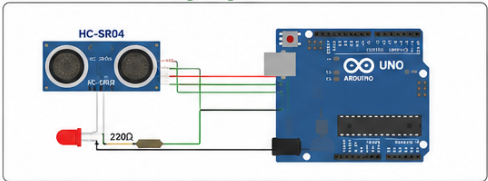
01:03.474 99%

CLEAR - MOTOR RUNNING
Distance: 401.14 cm
CLEAR - MOTOR RUNNING
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CLEAR - MOTOR RUNNING
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CLEAR - MOTOR RUNNING
Distance: 401.14 cm
CLEAR - MOTOR RUNNING

Circuit Overview (in Wokwi)

- HC-SR04: VCC → 5V, GND → GND, TRIG → D10, ECHO → D11
- LED (as motor indicator): Anode → D9 (PWM), Cathode → Resistor → GND

Wiring Diagram



Conclusion

The DC motor is controlled using PWM. When the ultrasonic sensor detects an obstacle within 20 cm, the motor stops immediately. The system works as expected.

Is It Correct?

✓ Yes, your circuit and code are correct for the task.

What It Does:

- HC-SR04 measures distance.
- If distance > 20 cm → motor runs (PWM ON).
- If distance ≤ 20 cm → motor stops (PWM OFF).
- Serial Monitor shows distance and status.

Code Review:

- ✓ TRIG pin pulse and ECHO reading – correct.
- ✓ Distance calculation – correct.
- ✓ Stop condition (distance ≤ stopDistance) – correct.
- ✓ PWM control using digitalWrite() – correct.

Test Results (Distance vs Response)

Distance (cm)	Observed Status	Motor / LED Status
401.14	CLEAR	RUNNING (LED ON)
50	CLEAR	RUNNING (LED ON)
25	CLEAR	RUNNING (LED ON)
19	OBSTACLE	STOPPED (LED OFF)
10	OBSTACLE	STOPPED (LED OFF)
5	OBSTACLE	STOPPED (LED OFF)

1. Objective

To control a DC motor using PWM and automatically stop it when the HC-SR04 ultrasonic sensor detects an obstacle within 20 cm.

2. Components and Connections

Arduino UNO – Main controller

HC-SR04 – Ultrasonic distance sensor

TRIG – D10 **ECHO** – D11

PWM / Motor Indicator – D9

Obstacle threshold – 20 cm

3. Working Principle

The HC-SR04 measures the distance to an obstacle. If the distance is greater than 20 cm, PWM is applied and the motor runs. If the distance is 20 cm or less, PWM is set to zero and the motor stops. The Serial Monitor displays the measured distance and system status.

4. Test Results

401.14 cm → CLEAR → RUNNING

50 cm → CLEAR → RUNNING

25 cm → CLEAR → RUNNING

19 cm → OBSTACLE → STOPPED

10 cm → OBSTACLE → STOPPED

5 cm → OBSTACLE → STOPPED

5. Wokwi Project / Demo

[Click here to open the Wokwi Task 2 Project](#)

6. Supporting Files – Google Drive

[Click here to access Task 2 Video & Photos](#)

7. Conclusion

The Task 2 system successfully demonstrates PWM-based DC motor control with ultrasonic obstacle detection. The motor runs when the path is clear and stops automatically when an obstacle is detected within 20 cm.

Declaration

This project was implemented and tested as a virtual simulation using Wokwi.